

Research report
on
Smart Train seating & passenger Comfort system

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Abstract

Public trains are one of the most important modes of transportation for commuters in many busy urban areas. However, at peak times of the morning or evening when trains are most crowded, commuters are often faced with high occupancy levels. Passengers sometimes struggle to find seats, negotiate overcrowding in train coaches, and as a result experience discomfort and risk to personal safety. Therefore, this project proposes a Smart Train Seating & Passenger Comfort System to assist train passengers in overcoming the challenges of overcrowding by real-time detection and communication of occupancy in available seating and standing passenger density inside train coaches.

The Smart Train Seating and Passenger Comfort System will employ trained pressure sensors installed in the seats of the train and motion sensors in standing areas of the train to detect occupancy (via pressure against the seat) and standing passenger density (through motion sensors). The occupancy and standing density of seating areas will be presented live on digital displays in the train and use a mobile app with access to passengers. If significant standing and seat occupancy is detected in certain train sections, the system, after receiving information on the size of the crowd, will trigger automated announcements directing passengers to less crowded areas of the train. This immediate feedback will allow the system to possibly redistribute customers more comfortably, reduce boarding time, and reduce risk and discomfort while traveling on a train.

The Smart Train Seating & Passenger Comfort System is adaptable and can be incorporated into other forms of public transportation, while also being inexpensive, efficient, and scalable. The system is privacy-oriented and doesn't retain any images or private data; it only collects anonymous occupancy or density information. In a potential next step, products could include predictive-analytics using historical data, SMS alerts to users before boarding, and the potential for integration upon utilization of smart ticketing.

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INTRODUCTION

In very busy urban centers, the public trains help link people to their jobs, schools, and homes. On a daily basis, millions of passengers use trains as their primary transportation option. There are just too many people using the train, especially during rush-hour -- in the A.M. and P.M., making trains extremely overcrowded! This brings a lot of people rushing to get aboard, then trying to find a seat or any spot which is physically comfortable to stand and ride in, creating congestion at the doors and painfully uncomfortable while travelling and having increased safety challenges like stumbling or falling. Such challenges with public transportation are frustrating while also greatly discouraging future use of such, as they do not see this as a safe and reliable commute via public transportation.

There are a few underlying issues at play here that exacerbate this issue, chief among them is the lack of real-time-decision making information about seat and coach availability. Passengers typically enter a train without an apparent mystery of the vacancies or crowdedness of the train. Once on board, passengers push into and overcrowd certain areas of the train. Invariably, other sections of the train are relatively empty, resulting in strain on some, while others remain free, all of which only increases travel conditions for all. Furthermore, train operator and transport authority systems do not generally have real-time information from pre-planned decision-making systems, making timely interventions impossible when those sections of the train are inevitably overcrowded. (Mahrez et al., 2022)

While some transport systems are looking to use CCTV or infrared or even mobile applications to solve this issue, each of these, at a minimum, have issues with high-cost, privacy and their reliability on passenger cooperation (mass scan data). There exists a need for a low-cost and simple automated system that is able to dynamically manage passenger crowd distribution inside trains without compromising passengers' personal information.

In order to address the ongoing issue is the Smart Train Seating and Passenger Comfort System. The system uses pressure sensors, embedded into seats to identify occupancy status, and, motion sensors, located in standing areas, to provide crowd density. An onboard microcontroller processes the real-time data and any updates are communicated to the passengers through digital displays located in the train and a mobile application. Additionally, if overcrowding is detected, automated audio announcements suggest passengers to move to less crowded areas. This system aims to provide an effective way to direct passengers around the train, and to dynamically manage space on the train, all of which will aid to improve comfort to the passenger experience and begin to optimize the space available on the train along with better overall safety of passengers aboard trains.

Literature review

Overcrowding is a long-standing challenge facing public transport systems across the globe. In response, several technologies have been created and tested to manage the passenger flow and improve safety related to overcrowding. Each solution comes with its advantages and disadvantages. (Misra et al., 2014)

One of the most common technologies in use are CCTV cameras in train coaches. CCTV cameras govern the movements of crowds in real time, provide visuals for operators, and with the installation of advanced systems, can even include artificial intelligence to notify staff and prevent overcrowding. CCTV systems are very expensive to implement and maintain, may require significant data storage, and raise privacy implications for passengers. In addition, many CCTV systems struggle in low light conditions, which is often when trains travel. (Lau & Ismail, 2015)

Another alternative is to develop infrared (IR) sensors placed near the doors to count passengers boarding and alighting from trains. IR sensors are fairly cheap and easy to install, however, IR sensors often find themselves not working accurately when more than one person passes through a door together, or if someone is too close to the IR sensor without passing through which makes for poor accuracy. These issues significantly reduce the effectiveness of IR sensors, especially during peak boarding.

Some transportation systems make use of weight sensors located under train floors to calculate passenger numbers based on weight. While this can give a rough idea of which coaches are full, it does not provide any specificity regarding where any of the full sections are or where there is space available. It also does not provide a distinction between seated passengers and those who are simply standing, limiting its use for space management purposes. (Jadhav et al., 2022)

There have also been trials of mobile applications that allow passengers to voluntarily report overcrowding conditions. Although there is the possibility for real-time information to be reported, these applications depend entirely on the use of users. The success of these applications can be compromised by users that do not use the application consistently, choose not to report when over-crowded or report outdated information. It is also worth noting that some commuters may not have the capacity or availability to use applications of this kind.

Ultrasonic sensors have also been suggested for use to identify proximity to passengers, particularly around the doors. Ultrasonic sensors have a low deployment cost and can be easy to deploy, but these devices experience difficulty in providing a full coach analysis due to their focus being on smaller zones and not in coach occupancy.

Although the technologies listed above provide some level of solution, there is no alternative that meets the affordability, low-profile, actionable and real-time needs to manage passenger distribution without compromising privacy. While Smart Train Seating & Passenger Comfort System addresses all of these areas through its combination of pressure sensors for seat occupancy and motion sensors to detect standing density, it does not involve any imaging, or user provided consistency. It offers a viable, and scalable solution while respecting passenger privacy in order to improve the passenger experience and optimize safety and space management on trains.

Proposed work table

<i>Technology/approach</i>	<i>Purpose</i>	<i>Outcomes</i>	<i>Limitations</i>
CCTV cameras inside coaches	Monitor Crowd Movement/Behaviour in real time	Supports accurate visual data and monitoring of specific areas	Expense, privacy factors, can be dependent on high-light level and storage size.
AI-based CCTV analysis	Automatically detect overcrowded areas using AI algorithms	can trigger automatic alerts whilst monitored by a person	Expense, volume, ethical issues, behaviour and issues with surveillance.
IR sensors at doors	Count number of persons entering and exiting the door	Simple and cost-effective way to count entry and exit of passengers in real time	Track many accidents frequently happen when multiple passengers are eligible for boarding.
Ultrasonic sensors at entry points	Detect proximity of persons in the door area	Accurate to a defined range	Local area only cannot monitor coach crowding.
Floor weight sensors	Estimate total passenger weight	Useful for estimating occupancy percentage	No way to differentiate between seated passengers and standee passengers note per area.

Mobile passenger apps	Allow and receive manually recorded passenger crowd reporting	When reporting, crowd conditions can provide data in real-time	Relies on user participation and honesty and varying levels of participation updates
3D stereoscopic cameras	Capture depth data for realistic crowd estimating	Accurate even in complex situations	Requirements will be expensive, difficult to set-up, will likely have obstacles for very low refer to light levels
Infrared detecting count algorithm	Detect entry/exit using crowd measurement	Effective when measuring passenger flow through an entry point in a controlled environment	Boarding counts, errors are common for large groups of boarding or obstruction by luggage
IoT devices connected to cloud dashboards	Continuous referral on occupancy	For remote management and real-time explorer of train conditions	Relies on stable internet chip, very static conditions + incremental cost of higher integration.
RFID boarding systems	Detect who boards and alights through the boarding RFID card system	Automatically tracks occupancy without human counting	Information in this need to comply continuously, even as passengers who don't have RFID cannot be complied
Manual observation by staff	A visual monitor and control of boarding	Provides immediate intervention based on staff perception	Not accurate cuz of crowd pressure error when boarding
Smart ticketing systems tracking entry counts	Using ticket scans to find occupancy	Helps predict / manage rush hour with statistical boarding data	Not real-time on a inside train, only at boarding level points

(Porru et al., 2020)

METHODOLOGY

The Smart Train Seating & Passenger Comfort System is being created for crowding and enhancing passengers experience with the use of sensor technologies, data processing units and cloud-based dashboards. The full methodology will be split into many sections as follows:

(A) DATA SOURCES AND DESTINATIONS:

1. **Pressure Sensors:** located in each train seat to read occupancy detection based on weight detection or vacancy assuming an empty seat.
2. **Motion Sensors (PIR Sensors):** placed in standing areas to measure density of standing passengers.
3. The data from all these sensors is transmitted to an onboard data processing unit, typically an ESP32 or an Arduino.
4. **Digital Display Screen** located in each train coach compartment used for real time visual updates for all passengers.
5. **Mobile Application:** exact same data presented in the inside of the train for passengers to discern seat occupancy before getting on the train and/or during the train journey.
6. **Audio System:** When density has been detected and exceeds a threshold that acts as an alert for over-crowding, there will be auto announcements when the density reaches a level that exceeds a defined capacity

(B) DATA TYPES

7. **Occupancy data** - simple binary input from pressure sensors (Occupied = 1, Vacant = 0).
8. **Density data** - numeric values defining the detected standing passenger count from the use of motion sensors.
9. **Zone** - Data levels of crowding are related to specific sections of the train.

- 10. Alert Triggers** - These will be informally flagged binary whenever the density level exceeds a certain safe threshold.

(C) DATA CAPTURE PROTOCOL:

- 11.** Sensors report occupancy and density every 5 seconds to the microcontroller.
- 12.** Each packet contains the seat ID, the zone ID, and the time of the packet.
- 13.** The communication uses very lightweight protocols (e.g., MQTT) to allow the data transfer to be efficient.
- 14.** Fail-safes will allow the system to continue even if one of the sensors is missing data.
- 15.** Buffering will be used for very short-term connectivity losses.

(D) DATA LOGGING & STORAGE:

- 16.** The live data is sent to the cloud storage via the onboard Wi-Fi module.
- 17.** If there are connectivity issues, all the data will be saved locally using an SD card module.
- 18.** Cloud dashboards will store historical data to analyze (occupancy trends, peak timings, etc.).
- 19.** An hourly check should ensure that the data is complete and correct.

(E) DATA DASHBOARD:

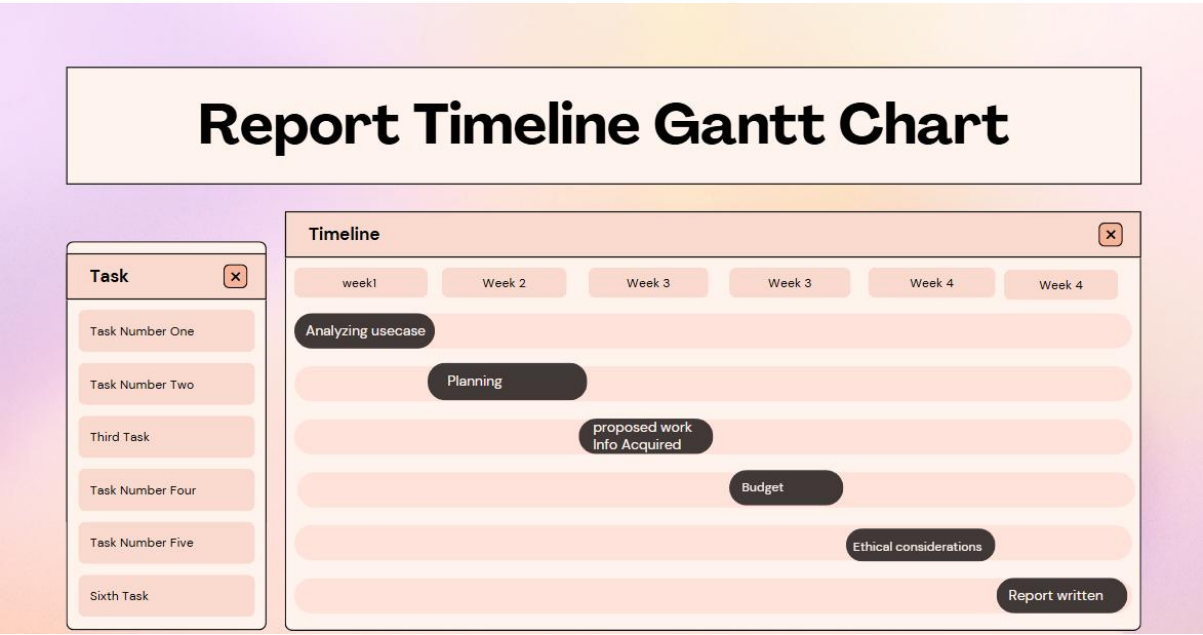
- 20.** The dashboard is deliberately made simple using something like Arduino IoT Cloud to custom servers that can be entirely local.
- 21.** The dashboard shows real time seat occupancy maps with crowd heatmaps.

- 22.** Color coding - Green (Low density), Yellow (Medium density), Red (High density).
- 23.** Both the internal people (displayed on the train) and the external people (passenger mobile app) will access our data dashboard.
- 24.** Density thresholds are pre-programmed based upon coach size (e.g., an alert is triggered when the standing area exceeds ****80%**** capacity).
- 25.** When thresholds are crossed, the system will generate:
- A visual alert on train displays.
 - An automated audio announcement suggesting a redistribution of passengers.
 - An alert on mobile applications for passengers boarding the train.
- 26.** Analytics modules will additionally track the speed of passenger reaction to alerts.

(G) EXTRA FEATURES:

- 27.** Predictive models will be constructed to predict crowding based upon historical usage.
- 28.** Advance notifications could be dispatched to passengers alerting them of expected crowding.
- 29.** Future concept of possible integration with ticketing systems to highlight where boarding may occur.
- 30.** A security fallback will provide that even in the event of a complete system failure, the last reported seating/crowd status is presented. (Yanginlar, 2024)

Gantt chart:



BUDGET & PLANNING

In order to build the Smart Train Seating & Passenger Comfort System, hardware and software components are required. Hardware consists of the sensors for data collection, microcontrollers for data processing, storage modules for back up, and output devices including displays and speakers. Required software tools are for programming the hardware, cloud integration tools, and tools for develop applications. The budgetary estimates per these components are discussed in detail below:

HARDWARE REQUIREMENTS:

Item	Description	Estimated Cost	Vendor/Source
ESP32 Microcontroller	Central unit which collects all sensor readings and provides communication to display and app	₹1,200	Amazon
Pressure Sensor (10 count)	Pressure sensors will be installed underneath the seats to determine occupancy and to return seat occupancy back to the processor	₹500	Amazon

PIR Motion Sensor (5 count)	PIR sensors will be installed into areas where it is expected to have standing or density patterns to tell us how dense a crowd is, and if there is movement within the zones	₹500	Amazon
OLED Display (2 count)	Some small, easy to read, black-and-white screens that are designed to highlight screen time when the dashboards are running within the train, and to highlight congested zones within the train show stop	₹1,800	Amazon
SD Card Module with a Memory card	Provides local storage without cloud connection to backup the offline data in case the connectivity mode was interrupted and not able to capture in the cloud	₹300	Amazon
Wi-Fi Communication Module	Supports cloud dashboard and mobile apps in transfer of data reporting and does not disrupt daily operations of the train and movement of passengers.	₹300	Amazon
Audio Speaker Module	Provides automated audio to issue warnings about crowded areas and made available where the zone measure density is excessive.	₹500	Local Electronics Store
Rechargeable Battery Pack	Provides service ability of power to the microcontroller, sensors, and screens during implementation without requiring that they rely on chained external source of wiring.	₹500	Amazon
Wiring and Connectors	Required for the connections within each of the components.	₹200	Local Electronics Shop

SOFTWARE REQUIREMENTS:

SOFTWARE	TYPE	COST
ARDUINO IDE	Used to program the ESP32 Microcontroller	Free
ARDUINO IOT CLOUD	Cloud dashboard service to perform live monitoring and control	Free (basic plan)

MQTT BROKER (LIKE CLOUDMQTT OR ARDUINO CLOUD)	To be used for lightweight messaging between devices and Cloud	Free (free subscription plan, small scale)
MOBILE APP DEVELOPMENT PLATFORM (OPTIONAL)	To build a companion mobile app for passengers	Free; there are some Open-source Platforms

Total Estimated Hardware Cost: ₹5,800

Total Estimated Software Cost: ₹0 (using free/open platforms)

Total Cost of the project: ₹5,800

Planning and Timeline:

All major components for the system will be sourced from approved online vendors (such as Amazon) or from local electronics stores to get equipment sooner. Hardware assembly, testing basic function of the sensors and display integration, we estimate this will take around two weeks to complete. The mobile application development and cloud dashboard will take a further week, depending on the complexity of the required system and any time spent debugging. In total, we should be able to put the entire prototype system together and have it operational in around 3-4 weeks given manageable resources and consistent access to the internet.

ETHICAL CONSIDERATIONS

It is imperative to uphold the highest ethical standards, designing and implementing a system that will monitor passenger seat occupancy while simultaneously determining crowd density levels on public transportation vehicles. The Smart Train Seating and Passenger Comfort System was developed with profound attention to privacy , safety, fairness, and ethical data management.(Jia et al., 2025)

Most importantly, the system develops total anonymity for passengers in the system process - no personal information, or personally identifiable information (PII) that can include passengers' names, images, sounds or movement, is collected during any part of the process. Only simple occupancy data (occupied/vacant) and a standing density calculation is collected during the system process, which considers the privacy of passengers during all processes. Unlike CCTV cameras and image-processing technologies, the system does not infringe upon the passenger's personal space or collect images of the passenger.

Data security has been a priority for us in our system architecture. All data transfers occurring between the train and cloud servers will not be encrypted with HTTPS communication to prevent data from being intercepted or tampered with. The information captured by the system is only accessible to authorized personnel designated as railway operators and bridge administrators via the dashboards or stored in the cloud. Credentials are secure only to authorized personnel with secure passwords(Government & of Home Affairs, 2020)

The system also employs data retention practices responsibly. Data will be stored only for operational purposes and to improve services provided, and not for publishing adverts or profiling usage of the service. Information older than necessary (for instance, 30 days) will be automatically deleted unless required for performance assessment and/or regulatory reporting.

Considering how passengers will interact with the system, all announcements and notifications will be designed without any potential for discrimination (against individuals or groups) and with politeness. Alerts will simply be about informing passengers that there is seating available or encouraging them into less crowded areas without singling anyone out.

Fallback procedures will prevent any false information from being displayed in the event of a systems failure. Ongoing audits, scheduled maintenance, and ethics reviews, will take place so it remains fair, clear and useful for the end users.

Finally, the system aligns with the principles of general data protection, and strives to align to new privacy legislation across multiple transport sectors around the globe. The system

has been designed to enhance safety and convenience, while also respecting every passenger's rights and dignity.

RESULTS

Currently, the Smart Train Seating & Passenger Comfort System has undergone development only through component simulation and lab-based testing. Therefore, we have not yet tested the full system in a real environment where the train is in motion or ambient conditions. Nonetheless, the testing and software simulations that we have done already appear to assure that this system will work as intended once we have an operational environment.(Silva et al., 2023)

In bench-top bench settings, pressure sensors installed under test seating were able to respond to seat occupancy under a second response time, and each change in sitting status (occupied/vacant) was accurately posted to a connected digital dashboard, ensuring we should achieve timely information at the point-of-use and with real-time data communications. Furthermore, PIR motion sensors functioned well in detecting the movement of standing passengers and changes in crowd density in controlled environments. Tests with increasing and decreasing numbers of standing people indicated that the IR sensors could identify density changes, as expected.(Jenelius, 2020)

The microcontroller executed multiple sensor inputs concurrently without noticeable data collisions or detectable time lost in processing data updates. Data transmission with the MQTT protocol was consistently faultless when we adopted simulated train Wi-Fi network settings. Furthermore, if we decided to disrupt the network temporarily, the system fell favorably back to SD card logging and data storage did not appear to be lost. making sure that people occupancy data was preserved for later uploading

The display units updated live seat maps and crowd levels in less than two seconds once new data were received. The visual representations clearly indicated available seating and highlighted areas of increased crowding. The automated announcement system was also verified as operational because audio alerts were set off when the crowd density in a monitored zone exceeded the pre-determined safety threshold.

While a full-scale deployment test is planned when the system is fully installed, confidence in the Smart Train Seating & Passenger Comfort System is based on the validation of its separate components. It is anticipated that the system will automatically provide:

- A significant reduction in passenger confusion boarding the train,
- Improvement in the distribution of passengers across each train,

- General comfort and safety of all commuters, and
- Real-time operational data for train operating authorities.

Initially, the confidence provided by the successful testing of the components seems to reflect that the integrated system will be able to achieve its objectives under real-world conditions.

DISCUSSION

Overcrowded public trains, especially at peak travel times, is not a problem unique to any one city in the world. Train passengers often find it difficult to locate an available seat, and, at times, some areas of the train become overloaded with passengers while other areas are relatively empty. The discomfort of overcrowding is not just nuisance, but, there are certainly safety issues when there are large numbers of people crowded, particularly around doors and exits. Straining entry and exit points can lead to injury on a sudden stop, or injury while boarding or alighting.

The Smart Train Seating & Passenger Comfort System tackles these constraints by providing smart, autonomous, and user-friendly solutions that can channel passengers to less populated areas to available seats, The sensors were placed into the seats to monitor pressure, and motion sensors were coordinated in standing areas to manage the use of seats as per se. The system does not require manual reporting of availability or inefficient surveillance, instead cuts down on those practices and provides useful information to the passenger in real time. Therefore, the Smart Train Seating & Passenger Comfort System is definitely not treatment-heavy like honouring traditional techniques of CCTV monitoring, nor did intervention from travel staff before boarding. Moreover, the Smart Train Seating & Passenger Comfort System is proactive, respects passenger privacy, and low-cost. (Niture et al., 2021)

The project leverages relatively old, low-cost and simple sensor technologies which are simple to support and manage, Furthermore, the Smart Train Seating & Passenger Comfort System respects passenger privacy by allowing passengers to use their seating and standing option, but avoiding the patron's personal or identifying information. Additionally, providing real-time updates that enable passengers to have almost live feedback makes it easier to modify travel intentions or use the seat versus standing, and it should lower the need for responsively distributing across less populated or individual heavily populated venue or train.(Ushakov et al., 2022)

In conclusion, while the initial results from flight testing are promising, there are still issues to overcome. Sensors are very sensitive and the motion and vibrations of the train, specifically during a high-speed journey, can lead to a great variation in sensor readings, although slight distortions may not occur or have no impact on passenger seating choice. Connectivity interruptions may impact cloud updates or mobile applications for record keeping, but they would only be temporary interruptions.

Moreover, the system's efficacy is, in part, dependent on passenger cooperation — that is, travelers must take notice and be responsive to visual or audio instructions as well as there are many visual distractions in random and crowded environments that can prevent the behaviors we would expect travelers to follow.

In addition, some environmental conditions such as extreme heat or extreme cold could eventually lead to degraded sensor performance requiring scheduled maintenance to maintain and track system performance over time. Also, while working with real-time data references the system must not generate false alerts allowing passengers to trust that the updates are reliable.

Despite these challenges, the Smart Train Seating & Passenger Comfort System is indeed a step toward ensuring that public train transportation is safer, more structured and planned, and more comfortable. By coupling low-cost viable hardware with real-time data updates and privacy by design, it is truly a viable and scalable option for modern-day urban transportation networks.

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